The isotropy-infimum core of Math427: angular-momentum block-diagonal

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

Since SC-SCOPE was lifted (2026-06-09), B1-RH-ENUM rests on the SOLE hypothesis H-LAYER, whose analytic core is Math427: within the diagonal-Gaussian class the isotropic dressing is the variational infimum. The companion note (hdiag-fullcovariance-formulation v1.0) proved the condensate-free functional F_0 convex; T-018 (hdiag-offdiag-gershgorin-threshold v1.1) reduced the residual off-diagonal obstruction to a single exchange scalar. This note completes the DIAGONAL half: it makes the isotropy of the infimum a STRICT, curvature-protected statement, exhibiting the exact mechanism by which rotation invariance forces the minimiser to be isotropic. Scope: F_0 over the diagonal-Gaussian class (H-diag), second cumulant, anchor $\mu^2 = 5 \times 10^{-3}$.

2. The condensate-free functional and its second variation

For a diagonal homogeneous Gaussian trial $G(q)$,

$$\frac{F_0[G]}{V} = \tfrac{1}{2}\int_q \ln G(q)^{-1} + \tfrac{1}{2}\int_q K_0(q)G(q) + \Phi_{\mathrm{diag}}(M_{\mathrm{tot}}), \quad K_0(q) = c(q^2 - q_0^2)^2, \quad M_{\mathrm{tot}} = \int_q G(q),$$

with $\Phi_{\mathrm{diag}}(M) = \frac{3u}{4}M^2 + \frac{5v}{2}M^3$ (Math427). The interaction depends on the trial state ONLY through the rotation-invariant scalar M_{tot}. The second variation at the isotropic dressing $G_*(q) = 1/(r_R + c(q^2 - q_0^2)^2)$ is

$$\delta^2 F_0 = \tfrac{1}{2}\int_q G_*(q)^{-2}\,\delta G(q)^2 + \Phi''_{\mathrm{diag}}(M_{\mathrm{tot}})\left(\delta M_{\mathrm{tot}}\right)^2, \qquad \delta M_{\mathrm{tot}} = \int_q \delta G(q),$$

entropy contributing $\frac{1}{2}G_*^{-2} > 0$ per mode, the kinetic term linear (no curvature), the interaction through δM_{tot} only.

3. Angular-momentum block-diagonalisation (the mechanism)

Expand $\delta G(q) = \sum_{l,m} a_{lm}(|q|)\,Y_{lm}(\hat{q})$. Because $\int Y_{lm}\,d\Omega = \sqrt{4\pi}\,\delta_{l0}\delta_{m0}$, ONLY the $l = 0$ component survives δM_{tot}. By orthonormality of $\{Y_{lm}\}$ the entropy quadratic form is already diagonal in (l, m), so the full Hessian is BLOCK-DIAGONAL in angular momentum. Writing the $l = 0$ radial mode $\delta G_{00}(q)$ with radial measure $d\mu(q) = |q|^2 d|q|/(2\pi)^3$, the interaction term is the explicit RANK-ONE positive form

$$\boxed{\delta^2 F_0[\delta G_{00}] = \tfrac{1}{2}\int_0^\infty G_*(q)^{-2}\,|\delta G_{00}(q)|^2\,d\mu(q) + \Phi''_{\mathrm{diag}}(M_R)\Big(\int_0^\infty \delta G_{00}(q)\,d\mu(q)\Big)^2, \quad H_{l\geq 1} = \tfrac{1}{2}G_*^{-2}}$$

i.e. the rotation-invariant interaction curves ONLY the isotropic breathing sector $l = 0$ (a rank-one positive operator built from the $M_{\rm tot}$ functional); every anisotropic sector $l \geq 1$ is interaction-FLAT.

4. Per-sector positivity (strict minimum, analytic)

(a) $l = 0$ **(breathing).** $\Phi''_{\rm diag}(M) = \frac{3}{2}(u + 10vM) = \frac{3}{2}u_{\rm eff}$; at the anchor $u_{\rm eff} = -0.86 + 10(3.24)(0.10941) = +2.685$, so $\Phi''_{\rm diag} = +4.03 > 0$: the rank-one interaction form is positive semidefinite, and the breathing sector retains in addition the strictly positive entropy term. **(b)** $l \geq 1$ **(anisotropic).** $H_{l\geq 1} = \frac{1}{2}G_*^{-2}$, with the ANALYTIC infimum

$$\tfrac{1}{2}G_*(q)^{-2} = \tfrac{1}{2}\left[r_R + c(q^2 - q_0^2)^2\right]^2 \;\geq\; \tfrac{1}{2}r_R^2 = 4.64 \times 10^{-2} > 0,$$

attained at the gap shell $|q| = q_0$. Positivity is therefore ANALYTIC (a square plus r_R^2), not a numerical grid value; the grid minimum equals $\frac{1}{2}r_R^2$ to $< 10^{-3}$. Both sectors are strictly positive, so G_* is a STRICT minimum of F_0 within the diagonal class: isotropy is protected by a positive curvature gap in EVERY angular-momentum sector. The mechanism is verified by exact Gauss–Legendre quadrature ($\int Y_{lm}\, d\Omega = 0$ for $l \in \{1, 2, 3, 4\}$ to 1.8×10^{-15}).

5. What remains (the honest residual)

This certifies the DIAGONAL half of Math427 (conditional on H-diag) as a strict, curvature-protected minimum, with analytic per-sector positivity. Two residuals remain, both mapped: (i) H-diag – the restriction to diagonal covariance – is Math427's standing hypothesis; (ii) the condensate-induced OFF-diagonal Hessian, which hdiag-fullcovariance-formulation isolated and T-018 reduced to the single competitor-dependent exchange scalar $b_{\rm exch}(Q) < b_*(Q)$ (binding full-class target 0.0206; the dressed Fock bubble, RES-5). Thus H-LAYER's analytic core now reads: the diagonal isotropy infimum is a STRICT minimum (this note, analytic), and the only obstruction to the full covariance is the exchange scalar (RES-5). No tier flip: B2 T6, B1 T6 on {H-LAYER}.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"The $l = 0$ interaction term was written as a scalar product $\Phi''_{\rm diag}\, w$."* ADDRESSED (operator review): §3 now writes it as the explicit RANK-ONE positive form $\Phi''_{\rm diag}(M_R)(\int \delta G_{00}\, d\mu)^2$ built from the $M_{\rm tot}$ functional; positive semidefinite since $\Phi''_{\rm diag} > 0$.

(β) *"The entropy floor is numerical."* RESOLVED (operator review): $\frac{1}{2}G_*^{-2} = \frac{1}{2}[r_R + c(q^2 - q_0^2)^2]^2 \geq \frac{1}{2}r_R^2 > 0$ is analytic, attained at $|q| = q_0$; the grid value 4.64×10^{-2} IS $\frac{1}{2}r_R^2$. The $l \geq 1$ protection is theorem-grade. (Headline tier kept T4 because the DIAGONAL restriction H-diag and the off-diagonal residual remain.)

(γ) *"Does this discharge H-LAYER or flip B1?"* DISMISSED: it strengthens the DIAGONAL core (conditional on H-diag) and leaves the off-diagonal exchange scalar (RES-5) as the residual. B2 stays T6, B1 stays T6 on {H-LAYER}.

Quantitative sanity check (mandatory): *mechanism* – $\Phi_{\rm diag}$ depends only on rotation-invariant $M_{\rm tot}$, Hessian on $l = 0$ alone ($\int Y_{l\geq 1} d\Omega = 1.8 \times 10^{-15}$); *sign* – bare $u = -0.86 < 0$ lifted to $u_{\rm eff} = +2.685 > 0$, $\Phi''_{\rm diag} = +4.03 > 0$; *entropy* – analytic $\geq \frac{1}{2}r_R^2 = 4.64 \times 10^{-2} > 0$; *consistency* – grid min $= \frac{1}{2}r_R^2$.

7. Result footer

Result ID: B2-PROPA-HLAYER / Math427 isotropy-infimum core (v1.1 rank-one + analytic-infimum re-issue)

Precise statement: Condensate-free F_0 over the diagonal-Gaussian class. The second variation at G_* is delta^2 F_0 = (1/2) int G_*^-2 (dG)^2 + Phi''_diag (dM_tot)^2. Since Phi_diag depends only on rotation-invariant M_tot=int G and int Y_lm dOmega=0 for l>=1, the Hessian is BLOCK-DIAGONAL in angular momentum: H_{l=0} = (1/2) int G_*^-2 |dG_00|^2 dmu + Phi''_diag(M_R) (int dG_00 dmu)^2 [rank-one positive]; H_{l>=1} = (1/2)G_*^-2 >= (1/2)rR^2 [analytic, attained at |q|=q0]. Both strictly positive (Phi''_diag=4.03>0; (1/2)rR^2=4.64e-2>0) => G_* is a STRICT diagonal minimum; isotropy is curvature-protected per sector, not fine-tuned.

Scope: Diagonal-Gaussian class (H-diag); second cumulant; anchor mu^2=0.005. Off-diagonal extension = T-018 exchange scalar.

Dependencies: Math427 (diagonal isotropy theorem); hdiag-fullcovariance-formulation (F_0 convexity, u_eff); hdiag-offdiag-gershgorin-threshold v1.1 (T-018 off-diagonal residual); Math424-AddA.

Evidence grade: ANALYTIC (exact block-diagonalisation + analytic entropy infimum (1/2)rR^2) + EXECUTED (res5_016_isotropy_infimum_core.py v1.1 6/6, GL exact quadrature).

Reproduction command: python codes/vacuum/res5_016_isotropy_infimum_core.py

Expected output: Phi''_diag=4.03>0; entropy infimum (1/2)rR^2=4.64e-2>0; <Y_{l>=1}>~2e-15; 6/6 PASS.

Falsification gate: An anisotropic (l>=1) diagonal perturbation that lowers F_0 (would require the interaction to curve l>=1, contradicting rotation invariance of M_tot); or Phi''_diag<=0 (u_eff<=0, refuted: u_eff=+2.685).

Tier before / after: No tier flip. B2 T6, B1 T6 on {H-LAYER}. Math427 diagonal core: strengthened from stationary-infimum to STRICT curvature-protected minimum, with ANALYTIC per-sector positivity (theorem-grade within the diagonal class).

No-overclaim statement: Certifies only the DIAGONAL half (conditional on H-diag) as a strict minimum. Does NOT discharge H-LAYER and does NOT flip B1/B2. The off-diagonal residual is the T-018 exchange scalar b_exch(Q)<b_star(Q) (binding 0.0206, RES-5). H-diag remains Math427's standing hypothesis.

Next required action: Bound the exchange scalar b_exch(Q) at operating intensity (RES-5/GAP-2) -- the single residual that, with this note and T-018, would extend the isotropy infimum from the diagonal class to the full covariance and discharge H-diag.